SSE

Malware Analysis Sandbox Booking

Full-stack | Concurrency | Time handling

Our products are built around cyber threat intelligence, so this problem is set in that domain. You do not need prior cybersecurity experience. A short primer is included below and you are welcome to use your AI assistant to learn the concepts. We will ask about the domain decisions you made in the interview.

Background

When analysts receive a suspicious file, they 'detonate' it inside an isolated sandbox: a disposable virtual machine where the malware can run safely while its behaviour is observed. Sandboxes are limited, and a sandbox must not be shared between two analysts at the same time or samples could contaminate each other. Every night each sandbox is reverted to a clean snapshot during a maintenance window.

The task

Build a full-stack system where analysts reserve sandboxes for a day.

Frontend

- A day view showing sandboxes as columns and time slots as rows, with existing bookings visible
- A form to book a sandbox (sandbox, date, start time, end time, analyst name, sample reference)
- A way to cancel a booking

Backend

- APIs to list sandboxes, view bookings for a date, create a booking and cancel a booking

Database

- Tables for sandboxes (seed 4-5, for example win10-sandbox-1, win11-sandbox-1, ubuntu-sandbox-1) and bookings

Business rules

- A sandbox cannot have overlapping bookings. Back-to-back bookings are allowed (for example 10:00-11:00 followed by 11:00-12:00).
- A single booking can be at most 2 hours
- No booking may overlap the daily maintenance window of 02:00-03:00 UTC
- Analysts work from different time zones, so be deliberate about how times are stored and displayed
- The system must stay correct when several analysts try to book the same sandbox and slot at the same time
- Return clear error messages for each type of rejection

Expectations

- Any language, framework or stack you are comfortable with.
- No authentication needed, and the UI can be plain.
- You are expected to use an AI coding assistant (Claude Code, Codex, Cursor, etc.). We are interested in how you use it.
- This problem is scoped to take about 20-30 minutes of focused build time with an AI assistant. Please do not spend more than [2 hours] in total, including the README and deployment steps. A working, well-understood core is better than extra features.
- Please submit within [X days] of receiving this assignment.

What to submit

1. GitHub repository (public, or private with access granted to [GitHub handle/email]) containing:

- A clear README covering: what the project does, tech stack, architecture and design decisions, how to run it locally, how to run tests (if any), API endpoints, known limitations, what you would improve with more time, and any security or domain assumptions you made
- Deployment instructions: step-by-step instructions to deploy the app (for example Docker/docker-compose, or a hosting platform of your choice), including required environment variables and how to seed or migrate the database
- A sensible commit history (please avoid a single giant commit)

2. Prompts used: the full list of prompts you gave to the AI tool (export, screenshots, or a markdown file in the repo is fine)

3. Build time: your start and end time for the build, and total time taken from first prompt to a working app

4. Tokens consumed: total input/output tokens used (for example the /cost command in Claude Code, the usage dashboard in Cursor, or the usage view in Codex/your account)

5. A short note (a few lines) on where the AI got something wrong or needed correction, and what you changed

Before you start

- Note your start time.
- Make sure your AI tool's usage or token tracking is available so you can report it afterwards.
- Create a new, empty GitHub repository.

Follow-up interview

Shortlisted candidates will be invited to a [30-45 min] interview. We will walk through your repository together, discuss your design decisions, trade-offs and domain understanding, and may ask you to explain or modify parts of the code. You should be able to explain any part of your submission, including code generated by AI.